Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B. Tech. EE Sem-VII Theory Examination

(March-2018)

EE402 Electrical Power system-IV

Date: 28/03/2018, Wednesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Figures to the right indicate full marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 **8**

Fig. 1

From Bus	To Bus	R (pu)	X (pu)	B/2 (pu)
1	2	0.05	0.15	0.1
1	3	0.04	0.25	0.05
2	4	0.05	0.15	0.08
3	4	0.05	0.20	0.06

For the system shown in Fig. 1, find bus admittance matrix (Y_{BUS}).

Q – 2 (A) Derive the static load flow equations for load flow analysis. Give the classification of buses and explain the importance of slack bus. **8**

OR

(A) Derive bus admittance matrix (YBus) by singular transformation method. **8**

Q – 2 (B) Explain different constraints in power system. **7**

OR

(B) Derive the equation for Penalty Factor. **7**

Q – 3 **Attempt any Two** **12**

(A) Define speed droop characteristic of generator with required diagram. Explain the Importance of keeping speed droop on synchronous generators operating in parallel.

(B) Draw the schematic for turbine speed governing system and explain the function of each component.

(C) In a system with two plants, the incremental fuel costs are given by,

$$\frac{\partial F_1}{\partial P_{G1}} = 20.0 + 0.010 P_{G1} \text{ Rs/h}$$

$$\frac{\partial F_2}{\partial P_{G2}} = 22.5 + 0.015 P_{G2} \text{ Rs/h}$$

The system is running under optimal schedule with $P_{G1} = P_{G2} = 100$ MW. If $\frac{\partial P_L}{\partial P_{G2}}=0.2$ find the plant penalty factors and $\frac{\partial P_L}{\partial P_{G1}}$.

SECTION – II

Q – 4 Formulate Z_{Bus} matrix for the system shown in fig.2; consider bus number four is a reference bus. **8**

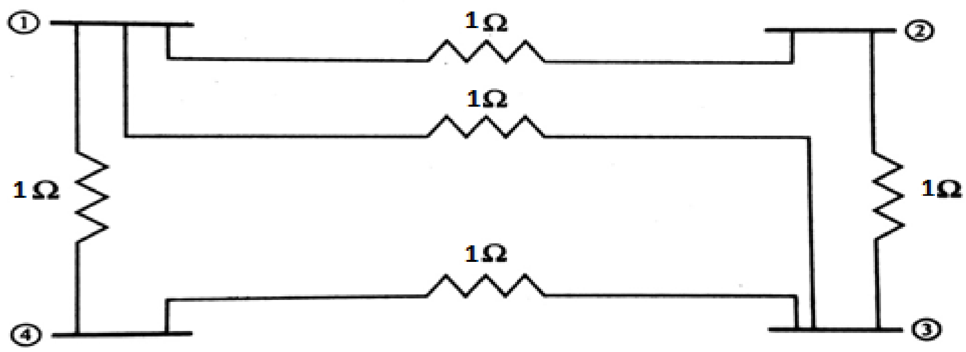

Fig. 2

Q – 5 (A) Derive the equation for formulating Z_{Bus} for Type-I and Type-II modification. **7**

OR

(A) Derive the equations of voltage and current for n bus system considering single line to ground fault. **7**

Q – 5 (B) Explain travelling wave on transmission line. Find expressions for surge impedance and wave velocity. **8**

OR

(B) Discuss the phenomina of wave reflection and wave refraction, hence derive the equation for Refraction and Reflection coefficient. **8**

Q – 6 **Attempt any Two** **12**

(A) A surge of 200 KV travelling on a line of surge impedance 400 Ω reaches a junction of the line with two branch lines of surge impedances 500 Ω and 300 Ω respectively. Find the surge voltage and current transmitted into each branch line. Also find the reflected surge voltage and current.

(B) How can bewley lattice diagram be drawn? Discuss its use.

(C) Compare the performance of GS, NR and FDLF load flow methods.

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